

Design Thinking for A.I. Solutions – Assignment 2

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Course: Design Thinking for A.I. Solutions

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Problem Subspace: Waste Classification System

Target Audience: Government Department

Part 1: Research Questions

- How can AI-powered waste detection systems in waste centers improve the efficiency and sustainability of urban waste management operations?
- What are the cost challenges faced by implementing smart waste collection and recycling process systems everywhere?
- What are the energy cost issues for running the system?

Part 2: Interview Questions

- Can you describe any current issues you face with waste classification when you throw away garbage?
- What features would make a smart waste bin system most useful for you or your home, and how much do you want to pay for a smart waste bin system?
- How do you perceive the benefits or limitations of smart waste bins?

Part 3: Database

1. OECD Environment Statistics

- a. The dataset displays greenhouse gas emissions from the plastic lifecycle, in million tonnes of CO₂ equivalent (MtCO₂e). The dataset provides information at the global level. Details can be found in the Annex of the OECD Global Plastics Outlook.
- b. URL: [Link](#)

2. World Bank – What a Waste Global Database

- a. What a Waste is a global project to aggregate data on solid waste management from around the world. This database features the statistics collected through the effort, covering nearly all countries and over 330 cities. The metrics included cover all steps from the waste management value chain, including waste generation, etc..
- b. URL: [Link](#)

3. Canada Waste Management Site

- a. The data class brings over data from the Waste Management Information System (WMIS) in Canada. This was married with the spatial data from Waste Disposal Sites, where possible Different Waste Disposal Site types collected by the Ministry of Natural Resources.
- b. URL: [Link](#)